

TVision

Smart Ad Reach Optimization

TVision x Rady School of Management, University of California San Diego

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Overview

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Group Member Introductions

Students in the UCSD Full-Time M.S. Business Analytics Program

- Program combines data science and business applications



Ruoqi Li

*Undergrad in
Applied Math and
Psychology*



Savitha Murali

*Undergrad in Electronics
and Communication
Engineering*



Emma Wu

*Undergrad in Fashion
Marketing, Finance and
International Business*



Erin Yabsley

*Undergrad in
Finance and
Computer Science*

Business Problem

Key takeaway: Advertisers need tools to measure incremental reach and reduce wasted spend from overlapping audiences across linear TV and CTV.



TV Audience Measurement
and
Analytics

Who is watching?

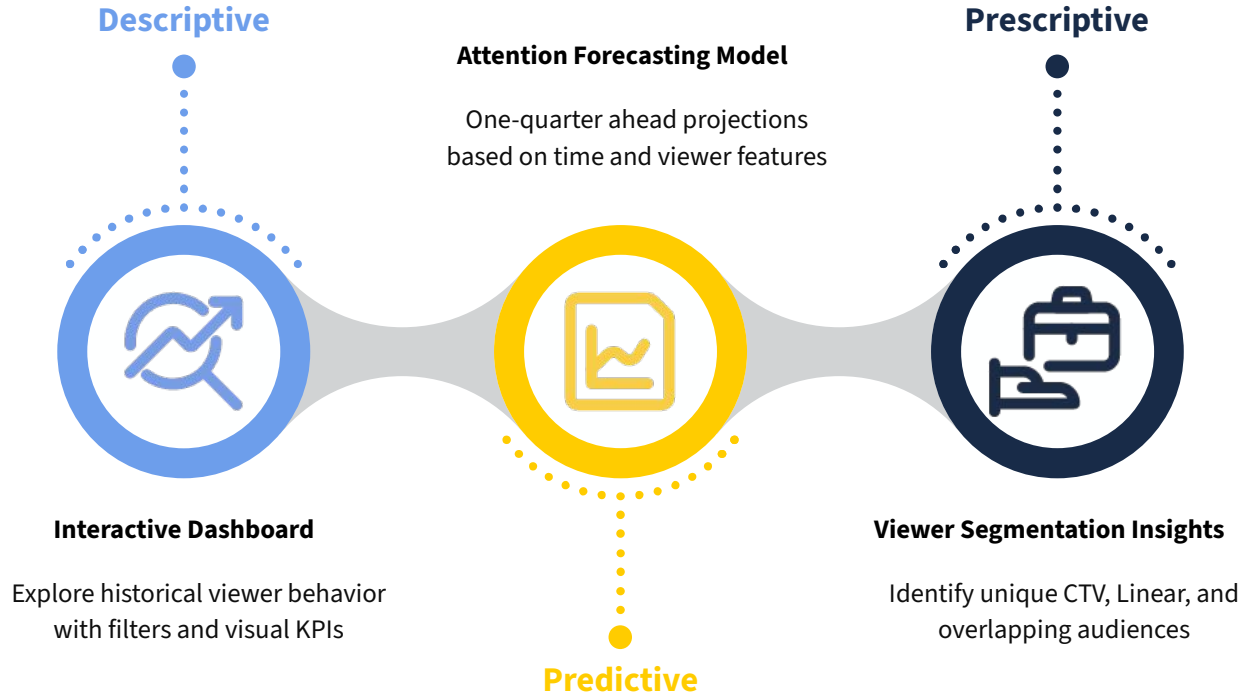
What are they watching?

Are they **attentive**?

Provide
advertisers with
clear visibility
into how their TV
ads perform
across linear TV
and CTV channels

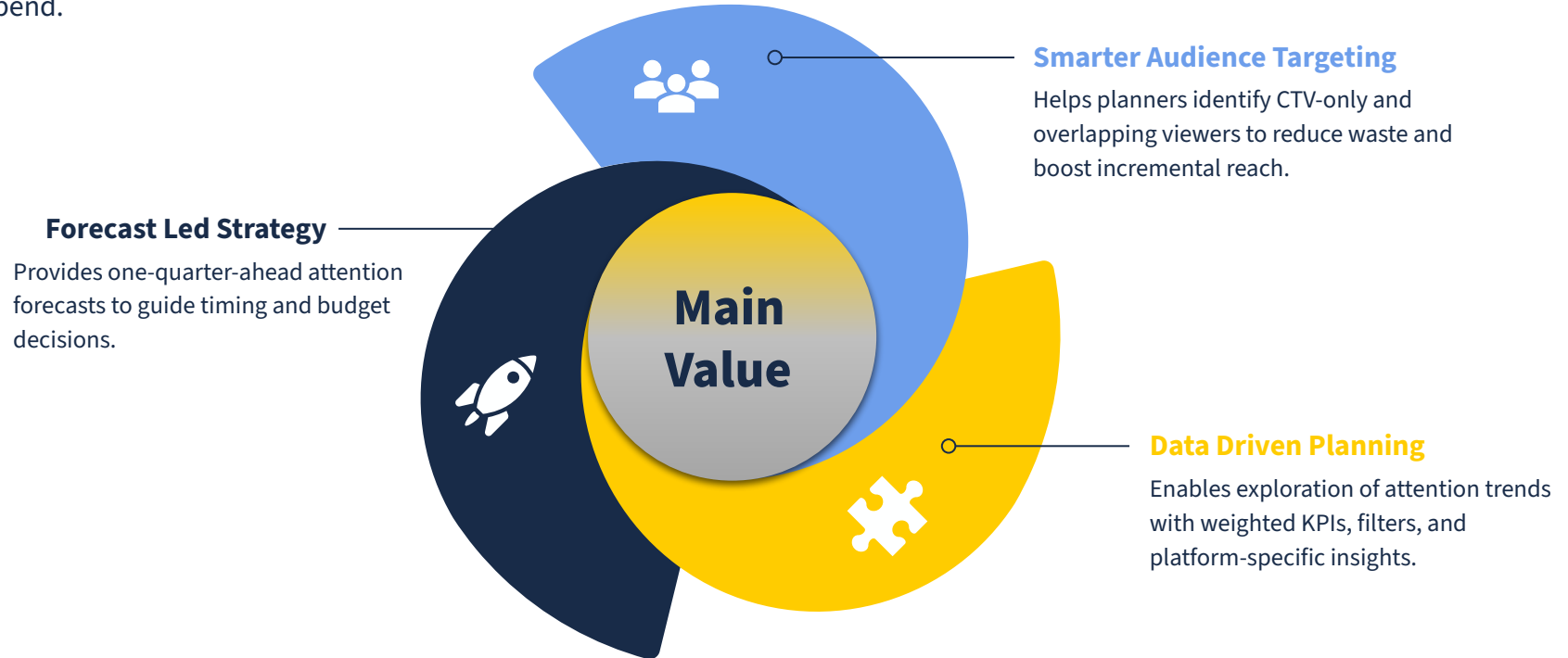
Deliverables

Key takeaway: We measure our success in our deliverables by measuring the deliverables' accuracy, flexibility, clarity, and actionability.



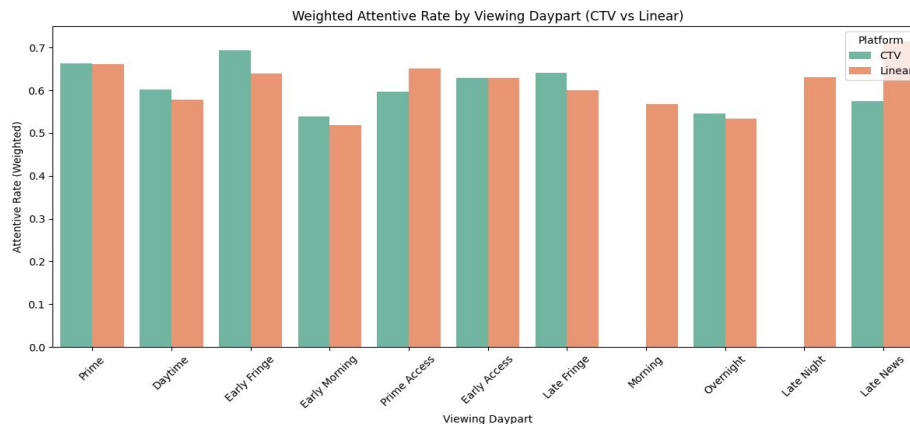
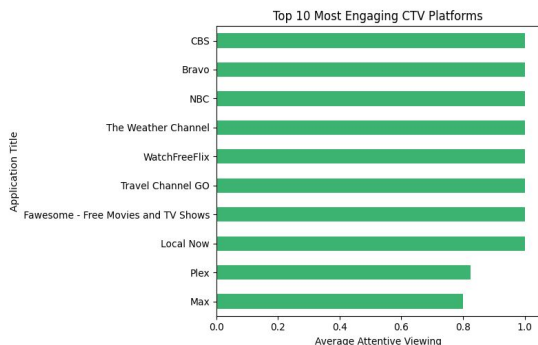
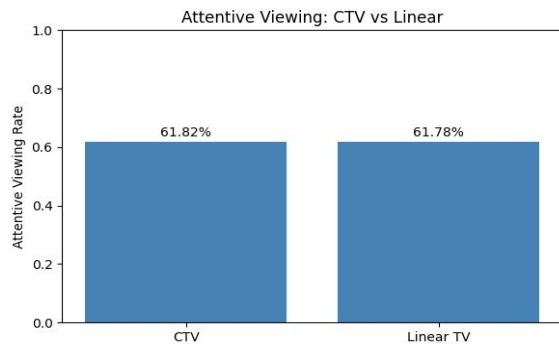
Business Impact

Key takeaway: Data-driven analyses provide value to TVision's clients and provides actionable insights for better ad spend.



EDA Summary

Key takeaway: Viewer attention varies significantly by daypart and platform, highlighting key opportunities for targeted media strategies.



The dataset includes **four quarters of data** for both **CTV** and **Linear TV** platforms:

- **CTV:** 5,812 records
- **Linear TV:** 68,421 records

Analytics Strategy

Key takeaway: Our model helps forecast viewer attention by platform and time block to support smarter planning.

Feature Engineering

Extracted the **month** variable

One-hot encoded key categorical features

Modeling Methods

ARIMA was used to model overall trends and seasonality.

SARIMAX added external factors like platform and daypart to improve forecast accuracy.

Model Selection

Models were evaluated using **AIC (Akaike Information Criterion)** and **BIC (Bayesian Information Criterion)** to ensure optimal balance between fit and complexity.

Analytics Results

Key takeaway: We forecast highest rates of attentiveness across different dayparts and platforms.

Channel Level Daypart-Level

ARIMA

Discovery+ and **Philo** showed the highest forecasted attentive rates across all platforms.

E! and **CMTV** are predicted to sustain the highest engagement levels over the next 3 months.

Prime and **Early Fringe** consistently outperformed other time blocks in predicted attentive rate

Daypart-Driven Attention Forecasting

SARIMAX

The **SARIMAX** model quantifies the impact of each daypart on attentive viewing.

Baseline forecasts suggest an attentive rate **around 0.64** under current scheduling.

When **Early Fringe** content share is **increased by 20%**, the model predicts a stable attentive rate near **0.65**, highlighting its strategic value for boosting engagement.

Platform × Daypart Forecasting

ARIMA

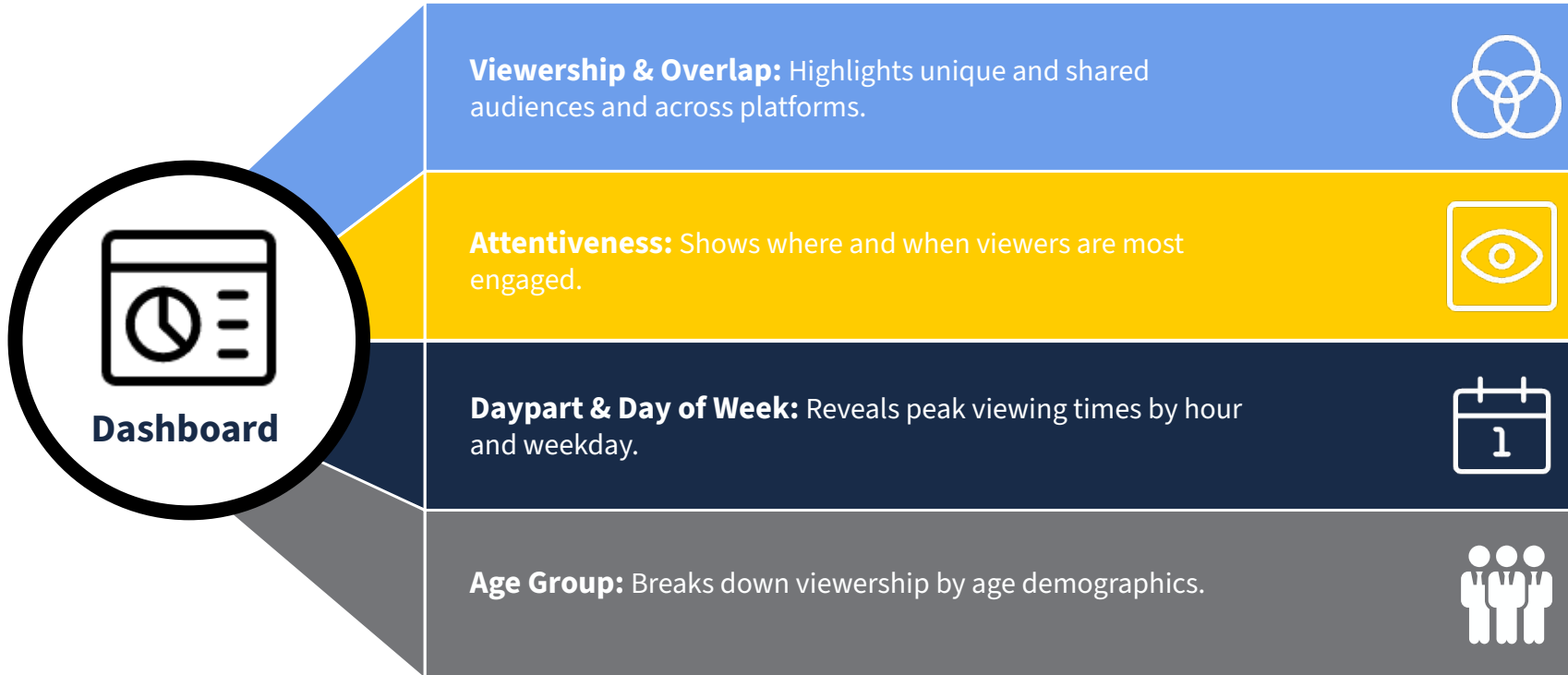
This model allows us to capture how audience engagement varies across **both time and platform.**

Amazon Prime Video in the **Daytime** block is projected to sustain high attentive rates over the next 3 months (**0.71–0.84**).

Some combinations show stable or **increasing** engagement such as **YouTube TV** in **Prime Access**, indicating strong alignment between content timing and audience behavior.

Deliverable

Key Takeaway: Different dashboard views present campaign-specific information that allow brand managers to understand the effectiveness of their ad reach.



Final Result

Key takeaway: We deliver value to TVision's end user by the dashboard's ability to filter on ad-specific campaigns and present viewership and attentiveness outcomes.

Ad IDs

Ad ID	Population
AE16546-2022-29-03955	51,198,985
AE16546-2022-36-03927	96,551,964
AE16546-2023-24-01669	12,697,966
AE16546-2023-27-03919	276,500,440
AE16546-2023-45-82062	134,551,323
AE16546-2023-51-02320	116,019,892
AE16546-2024-01-00642	327,504,573
AE16546-2024-11-08489	32,216,500
AE16546-2024-13-00966	68,923,670
AE16546-2024-13-04139	257,216,151
AE16546-2024-13-02844	6,337,921
AE16546-2024-23-01058	130,290,661
AE16546-2024-26-02504	50,184,015
AE16546-2024-33-02058	54,086,529
AE16546-2024-33-01489	50,488,768
AE16546-2024-35-03720	101,356,402
AE16546-2024-40-03481	20,578,866
AE16546-2024-48-01137	28,393,348
AE16546-2025-05-01177	7,434,205
AE16546-2025-07-02013	16,426,267

Viewers

Viewer	Population
Overlap	682,548,644
Unique CTV	35,913,951
Unique Linear	1,639,758,681

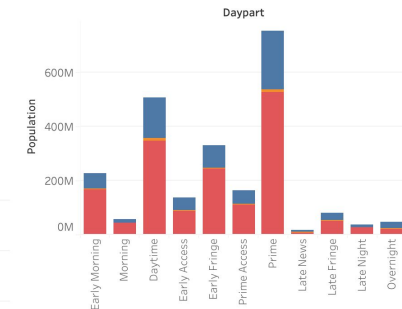
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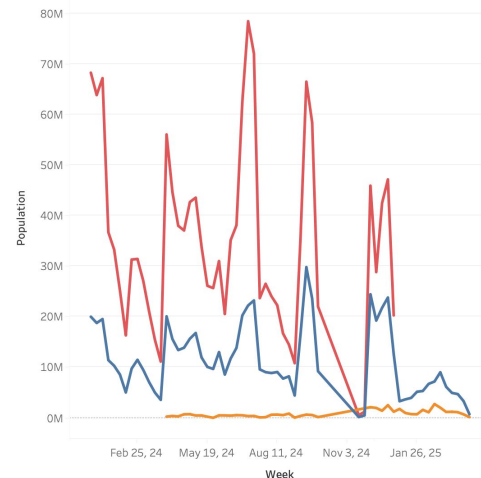
Overlapping Viewership

Overlap
Unique CTV
Unique Linear

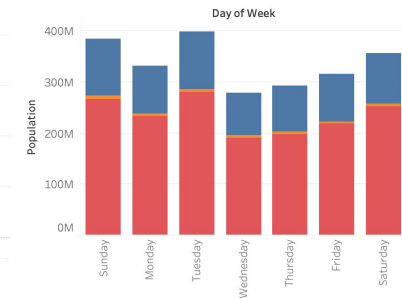
Exposure by Daypart & Viewer Types



Exposure by Week & Viewer Types



Exposure by Day of Week & Viewer Types

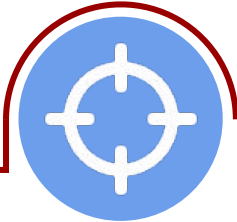


Conclusion

Key takeaway: We effectively met 3 of the 4 of our evaluation criteria, with accuracy lacking due to nature of advertising industry.

Accuracy

Industry metrics are challenging to predict and tuning the model did not result in significant improvement in predictive capabilities



Clarity

Dashboard presents information in digestible format and is in line with industry standards and needs of end users.



Actionability

Client confirmed the utility of dashboard for their end users (brand managers)



Flexibility

Provided scripts to allow new data to be input and reflected in the dashboard



Thank you!